

ACADEMIC MINOR PROJECT · PHASE 1

Student Retention Intelligence Platform

Phase 1 — Cloud Infrastructure Foundation on Microsoft Azure

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Based on SIH25102 — AI-Based Drop-out Prediction and Counseling System

One Platform, Built in Two Phases

PHASE 1 · THIS SEMESTER

Cloud Infrastructure Foundation

Automated, secure, reusable Azure infrastructure built with Terraform — Resource Groups, Networking, Storage, and Virtual Machines.

STATUS: IN PROGRESS



PHASE 2 · NEXT

AI Retention & Counselor Engine

Predicts dropout risk, explains why, recommends counselor action, and measures whether it worked — on top of Phase 1's infrastructure.

STATUS: PLANNED



Today's focus: everything in this deck is about Phase 1 only.

Infrastructure as Code (IaC) on Azure

Instead of manually clicking through the Azure Portal to create servers, storage, and networks, we describe the entire infrastructure in code (Terraform). Running that code builds — or rebuilds — the exact same environment, every time.



Repeatable

Same code → same infrastructure, every single time. No manual drift.



Secure by default

TLS 1.2+, no public storage access, no plaintext secrets — enforced in code.



Configurable via YAML

Change a setting file, not the code, to add or modify resources.

THE FLOW

YAML settings files
(what we want)



Terraform reads &
merges the YAML



Terraform modules
(how to build it)



Real Azure resources
(the result)

What Actually Gets Deployed on Azure

RESOURCE GROUP (rg-sample-dev-ae-01)



Virtual Network

1 VNet, address space 10.10.0.0/16

+



Subnet

"workload" subnet, 10.10.1.0/24

+



Storage Account

Blob containers + File shares

+



Virtual Machine

Windows VM + NIC + optional Public IP

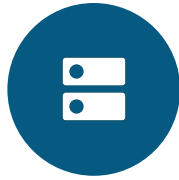
Everything inside the dashed box is provisioned automatically by running one Terraform command.

Four Reusable Building Blocks ("Modules")



resource_group

The container every other resource lives inside.



virtual_network

Network + subnets that resources connect through.



storage_account

Blob storage + file shares, with security defaults baked in.



virtual_machine

The compute — a Windows VM with NIC and optional public IP.

Each module is a self-contained folder with its own main.tf, variables.tf, and outputs.tf — like a LEGO brick that can be reused in other projects too.

Config Lives in YAML, Not in Code

Every resource has an `enabled: true/false` flag in a YAML settings file. Turning infrastructure on or off — or changing its name, size, or region — never requires touching Terraform code.



Right now: only the Resource Group is enabled — VNet, Storage, and VM are all switched off in settings, as a safe default.



Turning on a VM: set `enabled: true` on the VM, its VNet, and its subnet — Terraform figures out the rest automatically.

```
settings_virtual_machines.yaml
```

```
virtual_machines:  
  vm-sample-dev-ae-01:  
    enabled: false  
    resource_group_key:  
      "rg-sample-dev-ae-01"  
    vnet_key: "hub"  
    subnet_key: "workload"  
    size: "Standard_B2s"
```

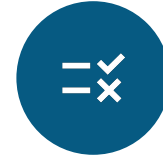
BUILT-IN FROM DAY ONE

Security Is Not an Afterthought



No public access

Storage accounts block public network access by default.



TLS 1.2 minimum

Hardcoded in the module — can't accidentally be weakened.



No plaintext secrets

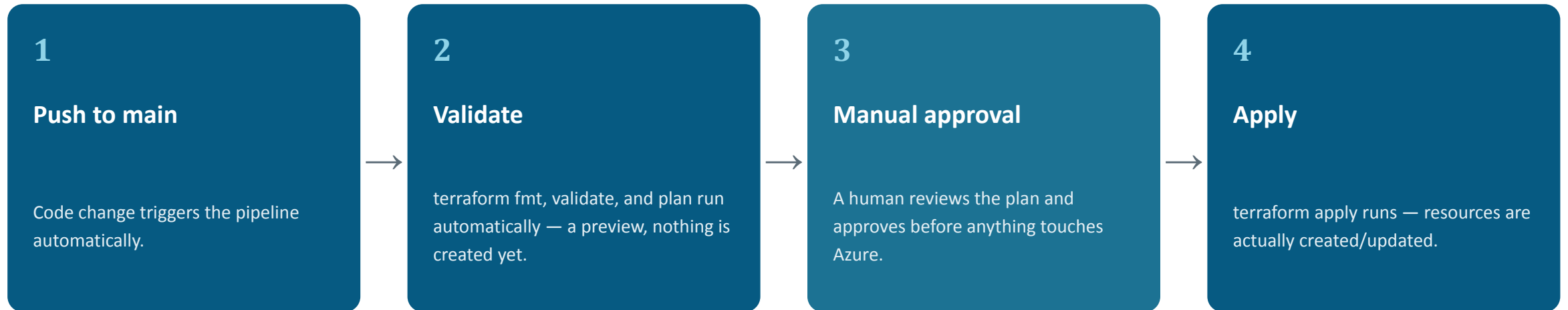
Designed for Key Vault / managed identity, not hardcoded passwords.



Predictable changes

terraform plan shows exactly what will change before it happens.

A Two-Stage, Human-Approved Pipeline



Manual approval is intentional — it prevents accidental changes to real cloud infrastructure.

WHY THIS MATTERS

A Reusable Foundation, Not a One-Off Build

This infrastructure isn't specific to our AI project — it's a generic, reusable pattern. Any future project needing Azure storage, compute, or networking can plug into the same modules.

WHAT WE STILL NEED TO DO THIS SEMESTER



Learn it deeply

Terraform, Azure resources, YAML pattern, CI/CD
— enough to explain it in viva.



Write the docs

Synopsis, SRS, and System Design diagrams —
currently only code exists.



Test & finalize

Deploy, test, integrate, and prepare the final
report, PPT, and demo.